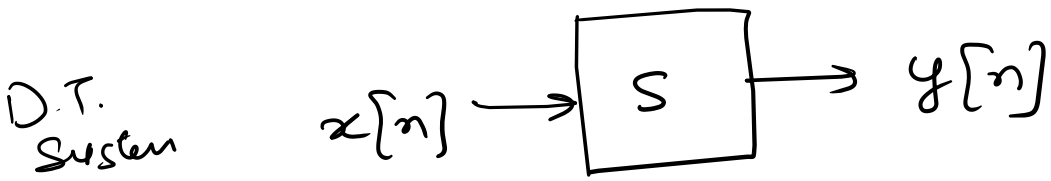
27/8/2025

Time - Invariance property of systems

D.T. System



$S: y[n] = S\{x[n]\}$

$x_1[n] \longrightarrow y_1[n]$

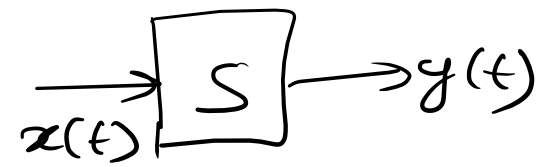
$\downarrow$

$y_1[n-n_0]$

$x_2[n] = x_1[n-n_0] \longrightarrow y_2[n]$

If $y_2[n] = y_1[n-n_0]$

Then it is time-invariant system



$S: y(t) = S\{x(t)\}$

$x_1(t) \longrightarrow y_1(t)$

$\downarrow$

$y_1(t-t_0)$

$x_2(t) = x_1(t-t_0) \longrightarrow y_2(t)$

If $y_2(t) = y_1(t-t_0)$

Then, the system is time-invariant

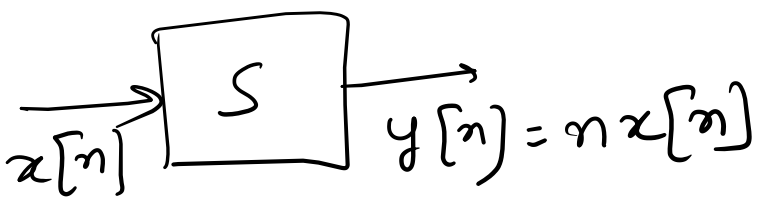
$$S: \quad y[n] = n\,x[n]$$

$$x_1[n] \longrightarrow y_1[n] = n\,x_1[n]$$

$$\downarrow$$

$$y_1[n-n_0] = (n-n_0)\,x_1[n-n_0]$$

The system is time-varying.



$$x_2[n] \longrightarrow y_2[n] = n\,x_2[n]$$

$$= n\,x_1[n-n_0]$$

$x_2[n] = x_1[n-n_0]$ IS $y_2[n] = y_1[n-n_0]$?

No

Linearity

If a system is linear, then it holds the properties of additivity & homogeneity.

Additivity :

$x_1(t) \longrightarrow y_1(t)$

$x_2(t) \longrightarrow y_2(t)$

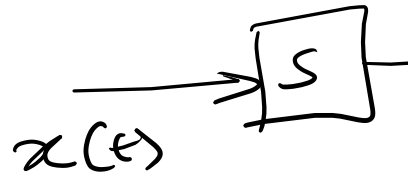
then $x_1(t) + x_2(t) \longrightarrow y_1(t) + y_2(t)$

Homogeneity

$\alpha x_1(t) \longrightarrow \alpha y_1(t)$

If the system is linear

$\alpha_1 x_1(t) + \alpha_2 x_2(t) \longrightarrow \alpha_1 y_1(t) + \alpha_2 y_2(t)$



$y(t) = 2x(t) + 3$

S: $y(t) = 2x(t) + 3$

$x_1(t) \longrightarrow y_1(t) = 2x_1(t) + 3$

$x_2(t) \longrightarrow y_2(t) = 2x_2(t) + 3$

$x(t) = \alpha_1 x_1(t) + \alpha_2 x_2(t) \longrightarrow y(t) = 2\{\alpha_1 x_1(t) + \alpha_2 x_2(t)\} + 3$

$= 2\alpha_1 x_1(t) + 2\alpha_2 x_2(t) + 3$

Compute $\alpha_1 y_1(t) + \alpha_2 y_2(t)$

$= \alpha_1 (2x_1(t) + 3) + \alpha_2 (2x_2(t) + 3)$

$= 2\alpha_1 x_1(t) + 3\alpha_1 + 2\alpha_2 x_2(t) + 3\alpha_2$

$\neq y(t)$

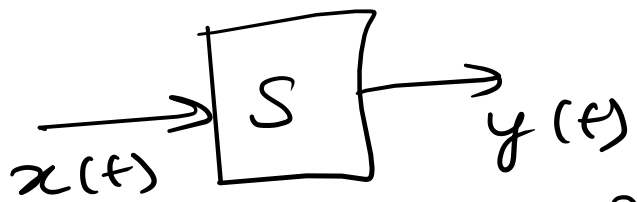
Hence, the system is not linear

i/p

If $y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$, then the system is linear

BIBO Stability

If For a bounded input, the system generates bounded o/p, then the system is called a stable system.



Same definition for D.T. systems

S: $y(t) = S\{x(t)\}$

Bounded i/p implies that $|x(t)| < M_x$ $\forall t$ where M_x is a finite constant

bounded o/p implies that $|y(t)| < M_y$ $\forall t$ where M_y is a finite constant

Ex $y[n] = \frac{1}{3}\{x[n-1] + x[n] + x[n+1]\}$

$y[0] = \frac{1}{3}\{x[-1] + x[0] + x[1]\}$

$y[1] = \frac{1}{3}\{x[0] + x[1] + x[2]\}$

⋮

If $|x[n]| < M_x$

then $|y[n]| < M_x$

This implies that system is stable.

Ex $y[n] = \sum_{k=-\infty}^{n} x[k]$

Let $x[n] = u[n]$

$y[-1] = 0$

$y[0] = 1$

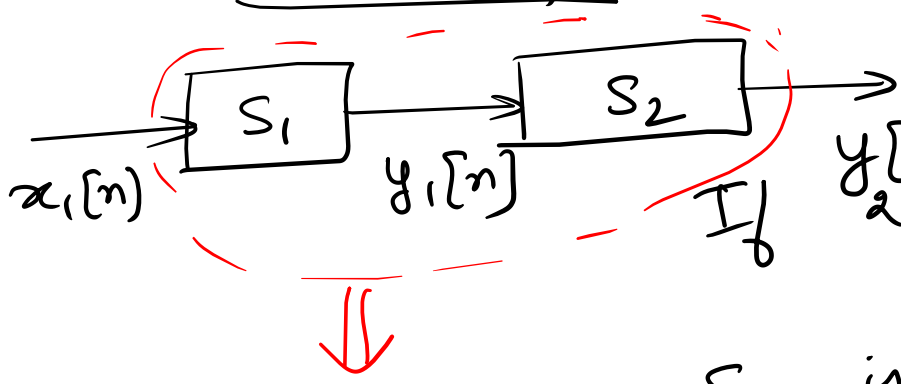
$y[1] = 2$

$$y[n] = (n+1)\,u[n]$$

Invertibility —

Same for C.T. system

D.T. system



If then we say that S_2 is inverse of S_1.

Identity system

S_1 & S_2 are connected in series or in cascade.

Example of C.T. system

S_1 : $y(t) = 2x(t)$

S_2 : $y(t) = \frac{x(t)}{2}$

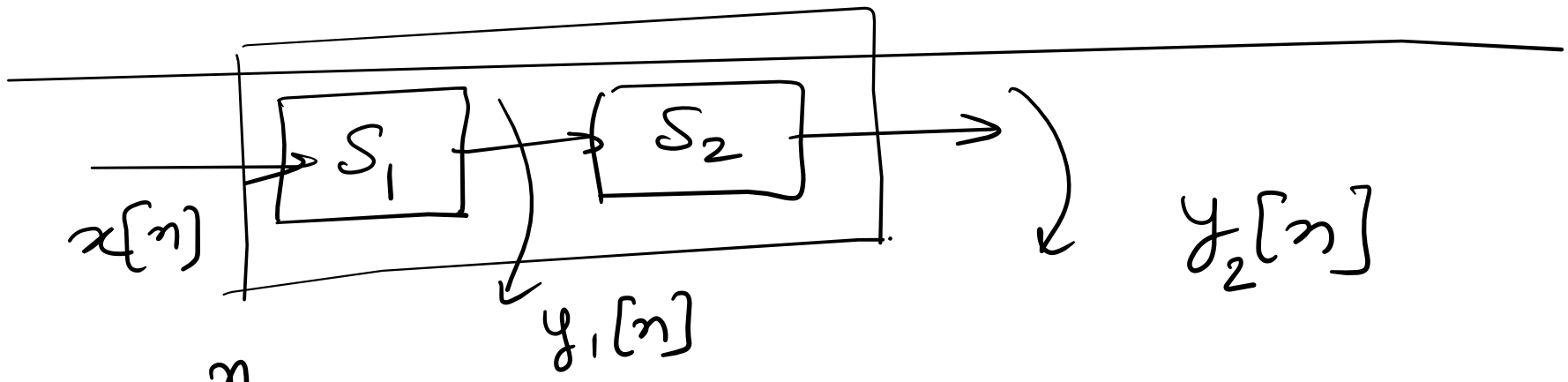
$y(t) = S_2 S_1 \{x(t)\}$

$= x(t)$

Q.1. System

$$S_1: \quad y_1[n] = \sum_{k=-\infty}^{n} x_1[k]$$

$$S_2: \quad y_2[n] = x_2[n] - x_2[n-1]$$



$$y_1[n] = \sum_{k=-\infty}^{n} x[k]$$

$$y_2[n] = y_1[n] - y_1[n-1] = \sum_{k=-\infty}^{n} x[k] - \sum_{k=-\infty}^{n-1} x[k]$$

$$= \sum_{k=-\infty}^{n-1} x[k] + x[n] - \sum_{k=-\infty}^{n-1} x[k]$$

$$= x[n]$$

1.14

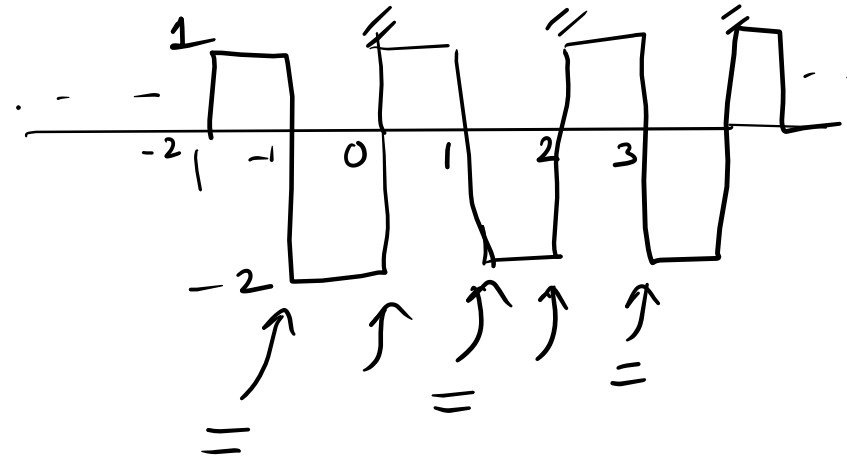
$$x(t) = \begin{cases} 1 & 0 \le t \le 1 \\ -2 & 1 < t < 2 \end{cases}$$

with period $T = 2$.

The derivative of this signal is related to

$$g(t) = \sum_{k=-\infty}^{\infty} \delta(t - 2k)$$

with period $T = 2$.



or can be shown that

$$\frac{dx(t)}{dt} = A_1 g(t - t_1) + A_2 g(t - t_2)$$

find A_1, A_2, t_1, t_2.